scientific reports

Check for updates

OPEN

Leveraging RegNet and CBAM for precise detection of honey adulteration using thermal image analysis

Boulbarj Ilias[1], Bouklouze Abdelaziz[2], En-Najy Anas[2], Samira Douzi[3]✉ & Hassan Douzi[1]

Honey adulteration poses a huge challenge with considerable health and economic consequences, underscoring the necessity for effective and precise quality evaluation techniques. This research introduces a novel approach for classifying levels of honey adulteration through thermal imaging and Artificial Intelligence (AI). Traditional detection methods are frequently marked by protracted processing durations, elevated expenses, and restricted sensitivity. To mitigate these constraints, a dataset of thermal images was compiled from 15 pure honey samples and 69 adulterated samples including glucose syrup at amounts between 1% and 30%. An adaptable AI model was created to categorize various honey types, attaining elevated accuracy, sensitivity, and specificity across different levels of adulteration. The model achieved a precision and specificity of 100% for pure honey and 1% adulteration, demonstrating strong performance at higher adulteration levels (0.98 and 0.97 for 3% and 5% adulteration, respectively). This methodology offers significant benefits, such as swift identification and versatility across various honey varieties. The results indicate that the integration of thermal imaging and AI can improve quality control in the honey sector, providing a dependable method for verifying the authenticity and safety of natural bee products. This approach facilitates enhanced quality assurance methods and bolsters consumer confidence in honey products.

Keywords Honey adulteration, Convolution block attention module, Attention mechanism, Thermal imaging, Quality control, Human health, Honey authenticity

Honey, also known as "liquid gold," has been highly valued for centuries, not only for its exceptional flavor but also for its versatile advantages. This material contains vital vitamins, minerals, antioxidants, and enzymes that have beneficial impacts on human health. Honey contains a high concentration of naturally occurring sugars, specifically fructose and glucose, which act as easily accessible sources of energy for the human body. Moreover, this remarkable natural material exhibits therapeutic, antioxidant, antibacterial, and anti-inflammatory properties, which hold promise for treating many skin conditions and infections[1].

Although highly regarded, the honey industry is currently experiencing a complex crisis caused by environmental pollution and the spread of diseases. This has resulted in an unstable situation where the supply and demand of honey are imbalanced, and sadly, has also led to instances of honey fraud. According to a recent field study, climate change has influenced the cognitive abilities of bees, specifically their learning and memory[2]. Bee populations have been adversely affected by environmental pollutants. Consequently, there has been a significant decline in the bee population, resulting in a substantial detriment to honey production. Simultaneously, the honey sector is confronted with the escalating spread of illnesses among bee colonies[3]. The combination of these two factors has resulted in a significant decrease in the availability of honey, which has failed to keep up with the increasing need for this organic sweetener. Consequently, the honey industry has created a fertile environment for deceptive activities, wherein authentic honey is adulterated with less expensive sweeteners or counterfeited using synthetic substances. Adulteration undermines both the quality and standards of the product, as well as jeopardizing consumer confidence and posing health risks.

Identifying honey adulteration is a complicated undertaking. In recent years, the focus of research has mostly been on instrumental analytical techniques, such as isotopic ratio analysis[4], chromatography[5], nuclear

[1]IRF-SIC Laboratory, Faculty of Sciences, Ibn Zohr University, Agadir, Morocco. [2]Biopharmaceutical and Toxicological Analysis Research Team, Laboratory of Pharmacology and Toxicology, Faculty of Medicine and Pharmacy, Mohammed V University, Rabat, Morocco. [3]Faculty of Medicine and Pharmacy, Mohammed V University, Rabat, Morocco. ✉email: s.douzi@um5r.ac.ma

magnetic resonance (NMR), and other spectroscopic methods like vibrational spectrometry (NIR, MIR, and Raman)[6–10]. Modern MIR and NIR instruments, in particular, allow for rapid analysis by obtaining a fingerprint of the samples in a few seconds without any need for sample pre-treatment. However, despite these advances, these sophisticated devices can still be expensive, limiting their accessibility. Therefore, there is a need for more affordable and efficient analytical techniques to detect adulteration[11].

Infrared thermography, or thermal imaging, is a method for observing and measuring the temperature of objects by detecting the infrared light they emit. This technology is very effective for examining surfaces or materials non-invasively, generating a "heat map" that displays temperature fluctuations. It can uncover or outline concealed attributes of goods, particularly gourmet items like honey. Thermographic cameras were initially employed as auxiliary instruments in medical applications[12], firefighting support, and military activities as sensors[13]. Within the food industry, this technology has been utilized to identify irregular elements throughout the complete process of producing various items and to oversee the standard of fruits in the agricultural field[14]. Thermographic cameras are gaining popularity in food analysis due to their non-destructive examination of samples, user-friendly operation, and rapid findings.

This study utilized thermographic cameras to record infrared emissions from honey samples with differing glucose syrup contents, with the objective of establishing a non-invasive technique for detecting adulteration. The thermal images obtained depict the mean temperatures of the sampled regions, highlighting variations in the components of the honey. Prior studies have shown that additions, such as sugar syrups, can modify essential thermal properties of pure honey, including the glass transition temperature[15,16]. Implementing an attention mechanism is key for efficiently extracting relevant information from thermographic images. This method concentrates the study on critical areas while reducing the intricacy linked to handling extensive datasets. Attention Mechanism[4] is a concept in deep learning that allows the model to focus on different parts of the input sequence when making predictions. The core idea is to give different "attention weights" to different parts of the input, so the model can selectively focus on the most relevant information for each step. Originally introduced in sequence-to-sequence models, attention has become fundamental for handling long dependencies in sequences by assigning varying levels of importance to different inputs. Models based on this mechanism have demonstrated their versatility by achieving exceptional results in various tasks in natural language processing and computer vision, including object recognition[17], video processing[18], image classification[19], unsupervised object discovery[20], unsupervised textual vision tasks[21], segmentation[22], and anomaly detection[23].

This study seeks to create a deep learning model that integrates a scalable convolutional backbone network[24,25] with an attention mechanism to examine thermographic images for detecting glucose syrup adulteration in honey. Thermographic imaging is very effective for this purpose, as it detects subtle temperature fluctuations that may indicate adulteration. The honey samples, sourced from various geographic regions and floral origins, offer a robust basis for evaluating the model's performance under different settings.

Our approach incorporates a scalable backbone architecture alongside an attention module to enhance feature extraction. The backbone network adaptively modifies its complexity, allowing it to discern both essential patterns and intricate correlations in thermal pictures. The attention module further augments this process by implementing two distinct forms of attention:

- Channel Attention emphasizes temperature-related attributes that aid in differentiating between genuine and contaminated honey samples by enhancing critical aspects linked to thermal variation.
- Spatial Attention concentrates on certain areas within each image, emphasizing locations that are most valuable for detecting evidence of adulteration.

The model is trained on a comprehensive collection of thermographic images featuring both genuine and counterfeit honey samples, enabling it to identify distinct patterns related to purity levels. Upon completion of training, the model may analyze new samples, serving as an accurate and dependable instrument for identifying adulteration.

This study enhances the utilization of deep learning and thermographic imaging in food quality assessment, providing a scientific basis for differentiating pure honey from counterfeit varieties. This methodology possesses considerable potential for practical applications in the food sector, enhancing the accuracy and efficacy of quality assurance methods.

This article is organized as follows: Sect. 2 delineates materials and methods, Sect. 3 elaborates on model development, Sect. 4 showcases results and discussion, and Sect. 5 concludes the entire paper.

Materials and methods

This section offers a detailed description of the honey samples used in the study, the acquisition and processing of thermal images, and the architecture of the proposed model.

Honey's samples preparation

The research amalgamated five honey varieties (Fig. 1) with glucose syrup as an adulterant to produce a series of tainted samples, as illustrated in Table 3. The emphasis was on two varieties of thyme, two varieties of euphorbia, and one variety of thistle. The honeys—Thyme-A, Thyme-B, Euphorbia-A, Euphorbia-B, and Thistle—were sourced from several places in Morocco, renowned for its production of thyme and euphorbia honey, between April and June 2023, assuring seasonal consistency (Table 1). The inclusion of numerous variants of each honey type facilitated the evaluation of the model's effectiveness in identifying adulteration and determining if geographical changes within the same honey type, across different adulteration degrees, influence model accuracy.

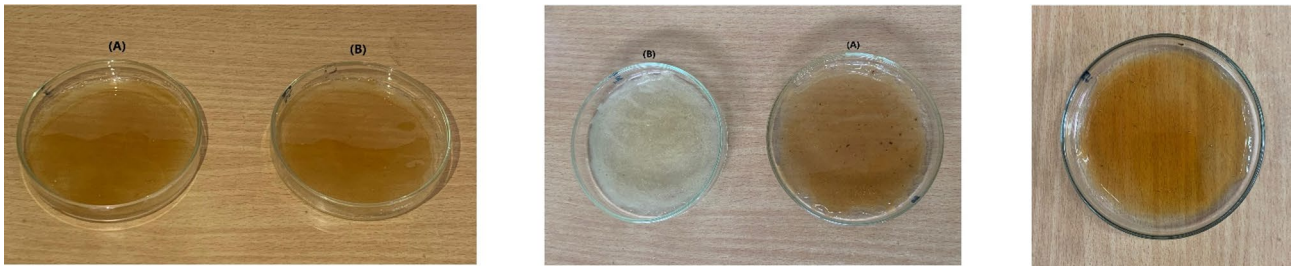


Fig. 1. The images of pure honey samples of different types.

Honey	Thyme-A	Thyme-B	Euphorbia-A	Euphorbia-B	Thistle
Province	Ida-Outanane	Taroudant	Assa-Zag	Agadir	Kénitra
Region	Souss-Massa-Drâa	Souss-Massa-Drâa	Guelmim-Es Smara	Souss-Massa-Drâa	Rabat-Salé-Kénitra
GPS Coordinate	31°06'06.4"N 9°39'58.4"W	30°40'38.6"N 8°10'41.9"W	28°30'15.3"N 10°31'39.4"W	30°46'04.5"N 8°21'03.4"W	34°14'24.1"N 6°35'41.8"W

Table 1. Geographic provenance of the honeys considered in this study.

Honey type	HMF (mg/kg)	Diastase activity	(% Brix)	Refractive index (20 °C)
Euphorbia- A	11.94 ± 0.17	17.6	80.568 ± 0.380	1.49219 ± 0.00099
Euphorbia- B	34.04 ± 0.22	11.5	81.951 ± 1.260	1.49585 ± 0.00334
Thyme-A	31.59 ± 1.18	25	82.162 ± 0.089	1.49683 ± 0.00090
Thyme-B	22.02 ± 0.34	15.8	80.486 ± 0.720	1.49602 ± 0.00039
Thistle	40.04 ± 0.31	21.4	81.572 ± 0.061	1.49342 ± 0.00111

Table 2. Physicochemical properties of pure honey samples.

Prior to thermal image acquisition, we examined the physicochemical characteristics of the pure honey samples, encompassing Hydroxymethylfurfural (HMF) content, diastase activity, Brix %, and refractive index. The HMF content and diastase activity were quantified utilizing the AOAC methods[26,27], thereby ensuring precision and reproducibility. Table 2 indicates that HMF level varied from 11.94 mg/kg to 40.04 mg/kg, keeping below the regulatory thresholds established by the European Union Directive (40 mg/kg) and the Codex Alimentarius (60 mg/kg). Diastase activity ranged from 11.5 to 25, adhering to worldwide honey quality criteria.

The experimental design comprised the preparation of 84 honey samples, partitioned into 56 samples for the training set and 28 samples for the test set, with each sample weighing 5 g (see Table 3 for distribution specifics). The adulterant utilized in this study was glucose syrup ($C_6H_{12}O_6$), duly sanctioned by the National Office for Food Safety for culinary uses. The syrup exhibited a dextrose equivalent (DE) between 40 and 60. The glucose was included into each honey sample at varying concentrations 1%, 3%, 5%, 10%, 20%, and 30% of the total sample weight to establish distinct levels of adulteration. Accurate measurements of these additions were obtained utilizing an analytical scale. To achieve uniform distribution, the glucose-honey solutions were thoroughly agitated until completely homogenous. Subsequent to preparation, the contaminated samples were incubated for 15 min at 60 °C to facilitate measurement. Thermal images were taken using an infrared camera oriented horizontally at a height of 25 cm above the samples, as illustrated in Fig. 2.

Thermal image acquisition

This work used a thermal camera to obtain images that illustrate the spatial distribution of thermal energy on the surface of heated honey by detecting its emitted thermal radiation. Fifteen-minute video recordings were produced to monitor and document the cooling process of each heated honey sample. The FLIR ONE PRO thermal camera runs at a frame rate of 8.7 Hz, within a spectral range of 8 to 14 µm, with an accuracy of 3%, a thermal resolution of 160 × 120 pixels, and a temperature detection range from 20 °C to 400 °C[12]. This portable device, which connects through a USB-C port, can be affixed to a smartphone. This study utilized different smartphones (Samsung S21 FE 5G[28] and LG Velvet 5G[29], with distinct configurations. The utilization of distinct devices aimed to improve the model's predicted accuracy in identifying new samples in real-world applications.

Thermal image preprocessing

Thermal image extraction

In this study, frames were extracted from videos of honey samples prepared for both training and testing purposes. Using FFMPEG software, frames were captured at 30-second intervals from the recorded sequences

Levels of Adulteration	Thyme-A	Thyme-B	Euphorbia-A	Euphorbia-B	Thistle	No. of samples
Training samples						
0%	2	2	2	2	2	10
1%	0	2	2	0	2	6
3%	0	2	2	0	2	6
5%	2	2	2	2	2	10
10%	2	2	2	2	2	10
20%	2	2	2	2	2	10
30%	2	0	2	0	0	4
Total	10	12	14	8	12	56
Test samples						
0%	1	1	1	1	1	5
1%	0	1	1	0	1	3
3%	0	1	1	0	1	3
5%	1	1	1	1	1	5
10%	1	1	1	1	1	5
20%	1	1	1	1	1	5
30%	1	0	1	0	0	2
Total	5	6	7	4	6	28

Table 3. Sample distribution by adulteration percentage for training and test Sets.



Fig. 2. Representative images of the thermal image acquisition process.

of each honey sample. FFMPEG was selected due to its versatility and robustness in video processing, making it well-suited for systematically extracting high-quality frames from thermal video footage[30].

The distribution of the extracted frames by adulteration level for the training and test sets is shown in Tables 4 and 5, as well as Figs. 3 and 4. During extraction, some frames exhibited imperfections, such as blurring and noise, which could compromise data quality and influence the study's outcomes. To maintain data integrity, frames with noticeable defects were excluded.

Region of interest detection

The subsequent phase entails a procedure referred to as Region of Interest (ROI) Detection, which focuses on pinpointing essential areas within each image. This strategy prioritizes the precise identification and delineation of vital areas to preserve all key information while eliminating irrelevant noise and superfluous features[31]. Picture processing techniques, including edge detection and picture segmentation, are employed to accurately isolate these important regions. The process initiates with the use of a specific filter to improve the original image's quality, succeeded by a pre-processing phase that transforms the image into grayscale. Enhancing image boundaries is essential for augmenting object visibility, hence enabling further analysis. The edge detection method discerns notable variations in light intensity throughout the image, consequently emphasizing outlines and object peripheries[32]. A contour map or mask is subsequently generated and superimposed onto the original

Levels of Adulteration	Thyme-A images	Thyme-B images	Euphorbia-A images	Euphorbia-B images	Thistle images	Count of Extracted Images by Concentration Level
0%	60	60	60	60	60	300
1%	-	60	60	-	60	180
3%	-	60	60	-	60	180
5%	54	60	60	60	60	294
10%	54	60	60	60	60	294
20%	60	60	60	60	60	300
30%	60	-	60	-	-	120
Number of Extracted Images by Honey Type	288	360	420	240	360	Total Number of Training Images **1668**

Table 4. Distribution of thermal images in the training Set.

Levels of Adulteration	Thyme-A images	Thyme-B images	Euphorbia-A images	Euphorbia-B images	Thistle images	Count of Extracted Images by Concentration Level
0%	7	-	-	12	30	49
1%	-	1	1	-	9	11
3%	-	18	-	-	30	48
5%	10	-	-	-	9	19
10%	11	17	2	3	8	41
20%	25	24	4	30	27	110
30%	21	-	20	-	-	41
Number of Extracted Images by Honey Type	74	60	27	45	113	Total Number of testing images 319

Table 5. The quantity of thermal images in the test set.

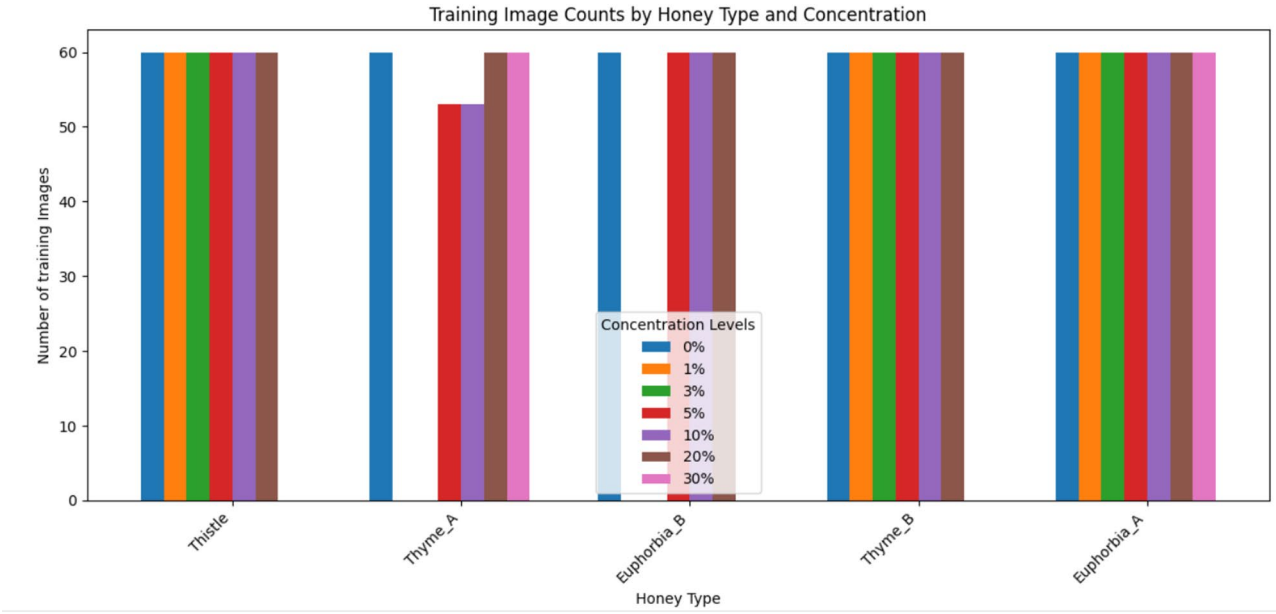


Fig. 3. Distribution of thermal images in the training set by adulteration concentration and floral origin.

image. This mask produces a revised image that carefully preserves essential information while excluding superfluous parts (refer to Fig. 5). This method guarantees that the analysis concentrates exclusively on the pertinent details critical for succeeding phases.

Images resizing

In order to maintain uniformity in the structure of the input data across the entire study, the image dimensions were standardized to 224 × 224 pixels. By employing this reduction technique, the capability of the classification

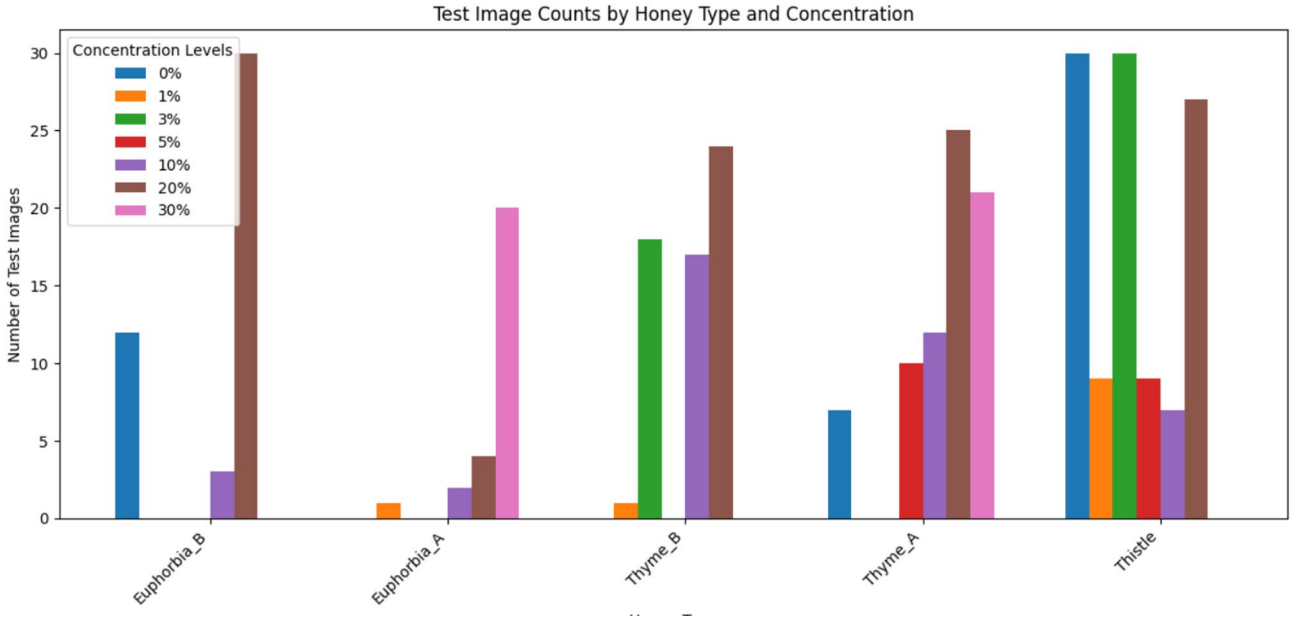


Fig. 4. Distribution of thermal images from the test set in percentage and floral origin.

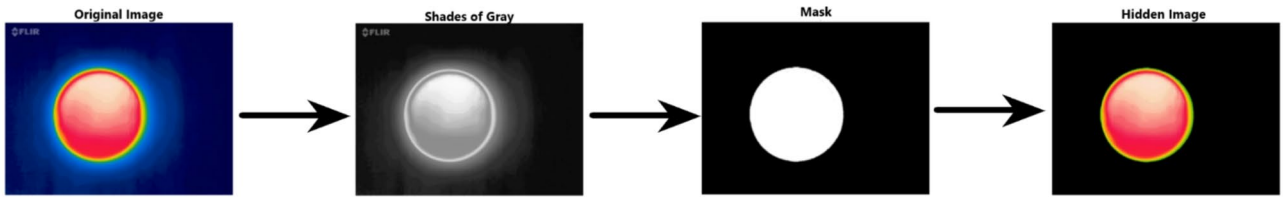


Fig. 5. Region of Importance Detection Process.

algorithm to prioritize the most prominent features in the images could potentially be improved. This is achieved by eliminating any superfluous backgrounds or irrelevant details that have the potential to consume larger images.

Models' development

Our model utilizes a dual-attention mechanism integrated with a convolutional neural network (CNN) architecture to enhance feature extraction from thermal pictures, specifically focusing on areas suggestive of honey adulteration. This method employs attention mechanisms in two dimensions-channel and spatial-to improve the features extracted by the CNN. This architecture enables the model to more efficiently discern pertinent patterns, highlighting both "which" aspects to prioritize (channel attention) and "where" to concentrate within each image (spatial attention).

Convolutional neural network (CNN) backbone

Our model leverages a convolutional neural network (CNN) architecture, serving as the primary feature extractor to identify essential patterns in thermal images, such as temperature distributions, textures, and structural details-key indicators of potential honey adulteration. To create an efficient CNN backbone, we implemented the Regulated Network (RegNet) architecture (Fig. 6), a state-of-the-art framework optimized for scalable and precise feature extraction, particularly suited for high-dimensional thermal imaging data[33].

A hallmark of RegNet is its systematic and regular design, which establishes uniformity across layers through a structured configuration of blocks. Unlike traditional CNNs that increase complexity by merely adding layers or expanding widths, RegNet's method scales depth, width, and complexity in a controlled and predictable manner, minimizing model variability and enhancing computational efficiency. This structured approach, organized into stages, captures progressively abstract features, allowing for incremental complexity while maintaining computational performance and naturally managing overfitting.

RegNet parameterized its network in a regular and interpretable way using four key parameters: w_0: The initial width (number of filters in the first layer), w_a: The width increment factor, controlling how the network grows, w_m: The width multiplier, influencing the number of filters in later layers and d: The depth, or the number of layers in the network. The width $w\,(i)$ of the i-th block is determined as:

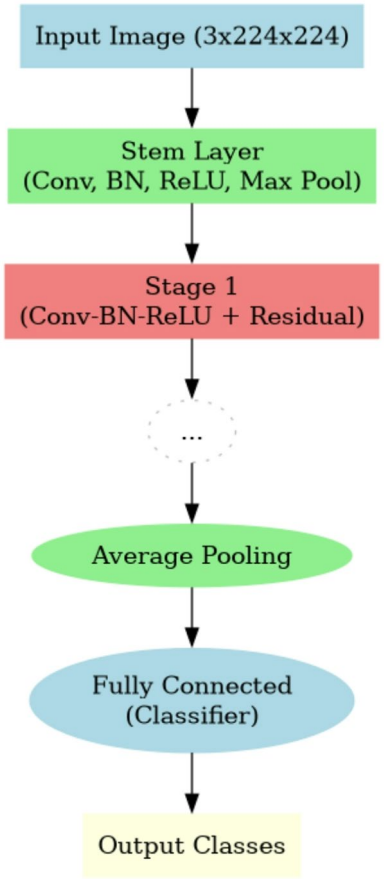


Fig. 6. Simplified RegNet Architecture.

$$w\,(i)\;=\;w_0\;+\;i\;?\,w_a \tag{1}$$

Where i ranges over the blocks in the network, and $w\,(i)$ is then multiplied by w_m to adjust the scaling of each block.

To enhance computational efficiency, RegNet employs bottleneck blocks, a type of residual block that reduces the number of operations and parameters while preserving representational power. A typical bottleneck block in RegNet can be mathematically expressed as:

$$y = x + Conv\,(BN(\,\sigma\;\;(Conv(\,BN\,(x)\,))\,))\,) \tag{2}$$

where:

- x represents the input tensor,
- BN denotes batch normalization[34],
- σ is the activation function (typically ReLU),

and Conv refers to convolution operations.

The residual connection x + y helps mitigate the vanishing gradient problem and enhances training stability by preserving information across layers.

Convolutional block attention module (CBAM)

The Convolutional Block Attention Module (CBAM) is a lightweight and effective attention mechanism that enhances feature representations in convolutional neural networks (CNNs) by focusing on informative regions in both spatial and channel dimensions. CBAM is composed of two sequential attention submodules: the channel attention module and the spatial attention module (See Fig. 7 (a)). First, the channel attention module operates on the feature map to refine the channels by emphasizing informative channels while suppressing less relevant ones. This is achieved by applying global average pooling and global max pooling across spatial dimensions (See Fig. 7), which capture different aspects of information, followed by two fully connected layers and a sigmoid activation function.

The output from the channel attention module is then multiplied by the original feature map to produce channel-refined features M_c. In short, the channel attention is computed as:

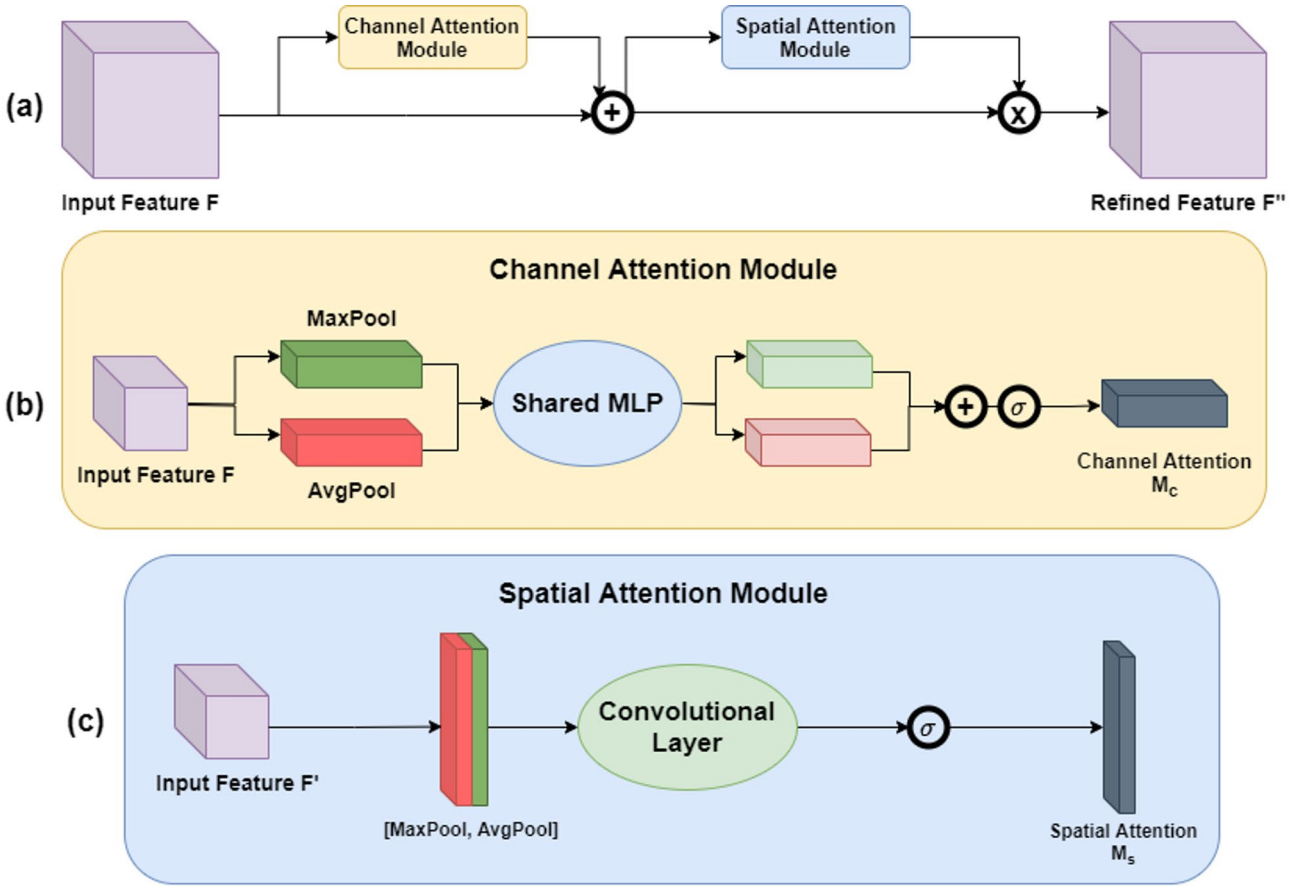


Fig. 7. CBAM Architecture.

$$M_c(F) = \sigma\ (MLP(\ AvgPool\,(F)) + MLP(MaxPool\,(F)\,)) \tag{3}$$

Where F is an intermediate feature map, the input of the CBAM (Fig. 7 (b)).

After channel attention, these refined features $F' = M_c(F) \otimes F$ are passed through the spatial attention module, which focuses on identifying important spatial locations within each channel. In this module, global average and max pooling are applied along the channel dimension to create two spatial descriptors that are then concatenated and convolved with a 7×7 convolutional layer to generate a spatial attention map $F'' = M_s(F') \otimes F'$. This map highlights essential spatial locations by assigning higher weights to informative areas, and it is element-wise multiplied with the channel-refined feature map to produce the final output (See Fig. 7 (c)). Through these dual-attention processes, CBAM effectively captures both 'what' and 'where' to focus on within the feature map, significantly improving the model's ability to focus on relevant features while maintaining computational efficiency.

In our research, CBAM was chosen for its dual-attention mechanism, which functions on both spatial and channel dimensions—essential attributes for assessing thermal images of honey samples. CBAM augments the network's ability to discern the most informative areas of an image by the successive application of channel and spatial attention. This approach allows the algorithm to identify nuanced patterns in temperature distributions and texture features that signify adulteration for honey quality assessment. The integration of CBAM enables the model to prioritize essential regions and features, thereby markedly improving the accuracy of honey adulteration detection by thermal imaging.

Pooling methods

The Global Average Pooling (GAP) layer is a key element in reducing the dimensions of feature maps after a sequence of CBAM is applied. Unlike a classic pooling layer (like MaxPooling) that simply selects the maximum values in a region, global Average Pooling takes the average of features maps (See Fig. 8). This allows for more global information to be maintained, which is less sensitive to local variations[35]. Assuming the size of the feature map is $h \times w$ with n channels, the Global Average Pooling of the k-th channel f_k is mathematically defined as follows:

$$f_k = \frac{1}{h \times w} \sum_{i=1}^{h} \sum_{j=1}^{w} X_{i,j,k} \tag{4}$$

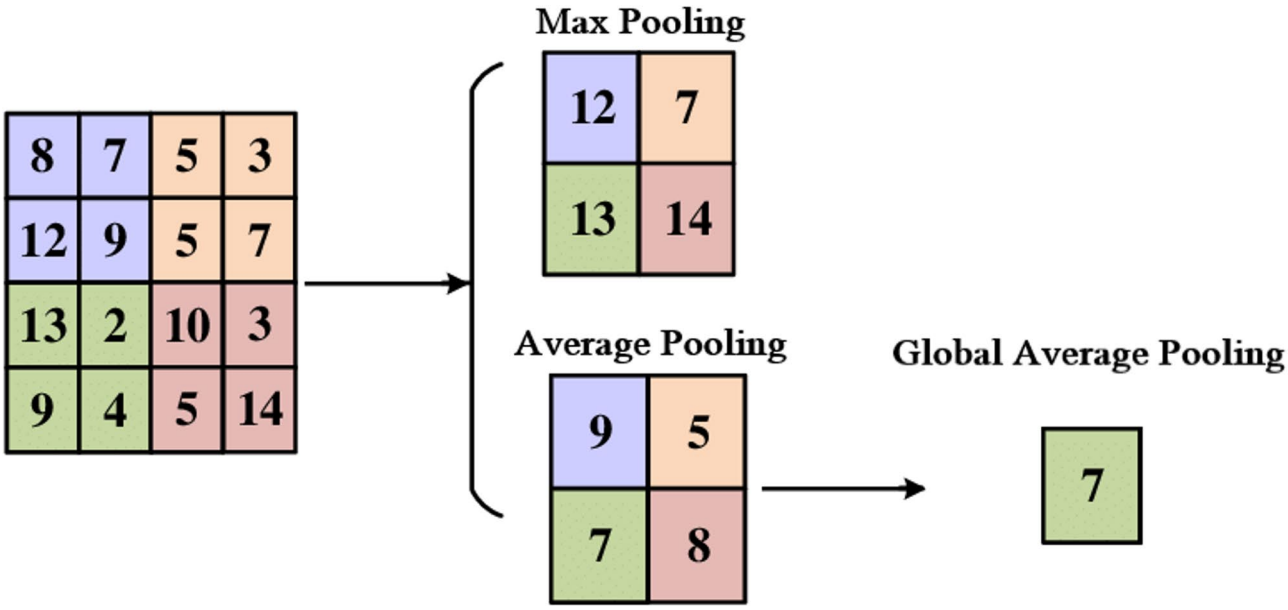


Fig. 8. Global Average Pooling.

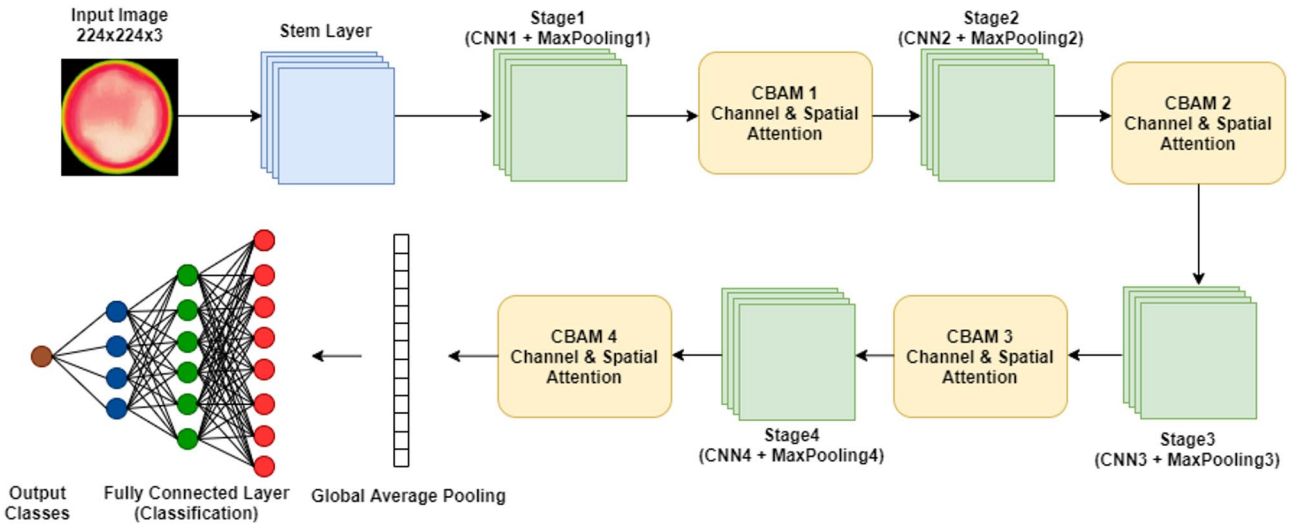


Fig. 9. Proposed model architecture.

Where $X_{i,j,k}$ is the value of the feature map at the position (i, j) in the k-th channel. The main advantage of GAP is that it reduces the size of the feature maps without significant loss of information, while limiting the risk of overfitting. GAP will allow us to compact the thermal information captured in each image while maintaining a rich and discriminant representation. In addition, this reduction of dimensions facilitates the transfer of these characteristics to the final classifier, producing a reduced and optimized vector of characteristics for classification.

Flow of thermal image data through model components

The proposed approach (Fig. 9) utilizes thermal images of honey, which reflect temperature distribution differences linked to varying amounts of adulteration, as input with dimensions ($224 \times 224 \times 3$). Every input image is an RGB color image, characterized by three channels. The model's initial layer, known as the "stem layer," functions as the primary processing element within the RegNet backbone, executing an initial convolution to produce a feature map. This feature map encodes fundamental patterns, such as edges and textures, which constitute the essential components for advanced analysis in later phases.

The output from the stem layer is transmitted directly to the initial processing stage, bypassing the Convolutional Block Attention Module (CBAM), hence facilitating the efficient management of low-level characteristics. The model design utilizes four blocks of convolutional layers, each succeeded by max pooling

Hyperparameter	Batch size	Number of epochs	Learning rate	Optimizer	Loss function	Reduction ratio (CBAM)	Input image size
Value	32	100	0.001	Adam	Cross Entropy Loss	16	224 × 224

Table 6. Hyperparameters of the model.



Fig. 10. Study workflow overview.

(CNN + MaxPooling). The convolutional blocks systematically extract progressively intricate hierarchical characteristics from the thermal images, augmenting the model's comprehension of spatial and contextual information. At each phase, spatial dimensions are diminished, enabling the model to concentrate on the most pertinent features.

Subsequent to each CNN + MaxPooling block, the architecture incorporates the Convolutional Block Attention Module (CBAM), which augments feature representation by implementing attention in both "channel" and "spatial" dimensions. This attention module utilized four times (CBAM 1, 2, 3, and 4), progressively enhances the feature maps to augment the model's emphasis on essential details.

Subsequent to feature extraction, Global Average Pooling is implemented on each feature map, condensing it to a singular value by averaging over all spatial regions. This phase converts the 3D feature maps into a 1D vector, markedly decreasing data dimensionality and priming it for classification. The fully connected layer subsequently translates the 1D vector into output classes, denoting varying degrees of honey adulteration, from 0% to 30%.

We utilized the hyperparameters listed in Table 6 to train our model. The model underwent training for 100 epochs with a batch size of 32, employing the Adam optimizer with a learning rate of 0.001 to minimize cross-entropy loss. Input images were scaled to 224 × 224 pixels, and the model categorized the data into seven unique classes. Channel Attention and Spatial Attention (CBAM) modules with a reduction ratio of 16 were integrated to improve feature extraction. Training was performed on a GPU to enhance computational performance. The hyperparameters were chosen to optimize performance and training efficiency, hence facilitating robust learning and generalization throughout the dataset.

Results and discussion

This section assesses the efficacy of the proposed model, which employs thermal imaging of five honey varieties - Euphorbia_A, Euphorbia_B, Thyme_A, Thyme_B, and Thistle—to classify samples across different adulteration degrees, ranging from pure (0%) to significantly contaminated (30%). As illustrated below in Fig. 10, the study workflow encompasses four key stages: data collection, image processing, model development, and evaluation. The model underwent training and evaluation using a comprehensive dataset of 1,668 thermal pictures depicting these particular honey kinds.

The training and validation curves (Figs. 11 and 12) indicate that the model is acquiring knowledge efficiently, as both losses markedly decline throughout the initial epochs. The training accuracy approaches 100%, however the validation accuracy is marginally lower and exhibits considerable variability. The validation loss displays fluctuations, indicating that the model's performance on unseen data is marginally less steady than on the training set. The overall trend signifies robust learning, as the validation accuracy roughly aligns with the training accuracy, indicating effective generalization of the model to the validation data.

The evaluation metrics and confusion matrix shown in Table 7; Fig. 13, highlight the model's efficacy in identifying honey adulteration levels, especially for pure samples (0%) and high adulteration levels (20% and 30%). The precision values for these categories are remarkably high, with scores of 1.00 for pure and 1% samples, and 0.85 or higher for 20% and 30%, demonstrating the model's proficiency in reliably classifying these levels with little false positives. Correspondingly, recall values are strong for these categories, attaining 0.94 for pure samples and surpassing 0.90 for 20% and 30% adulteration levels. This demonstrates the model's ability to accurately classify the majority of data within these categories. The F1 scores exhibit a continuous trend, affirming reliable performance in balancing precision and memory for these essential areas.

The confusion matrix further emphasizes the model's efficacy in differentiating between pure and significantly contaminated samples. Significantly, there are no occurrences of adulterated samples (≥ 1%) being erroneously classified as pure (0%), illustrating the model's dependability in detecting contaminated samples, an essential criterion for quality control. Moreover, categories exhibiting modest adulteration, specifically 1% and 3%, get superior performance, with precision, recall, and F1 scores over 0.90, highlighting the model's ability to identify even slight degrees of adulteration.

Notwithstanding the model's robust overall performance, certain flaws manifest, especially in its capacity to recognize mid-range adulteration levels (5% and 10%). The precision levels are diminished (0.57 and 0.72, respectively) and recall is also lower (0.63 and 0.68, respectively), resulting in decreased F1 scores (0.60 for

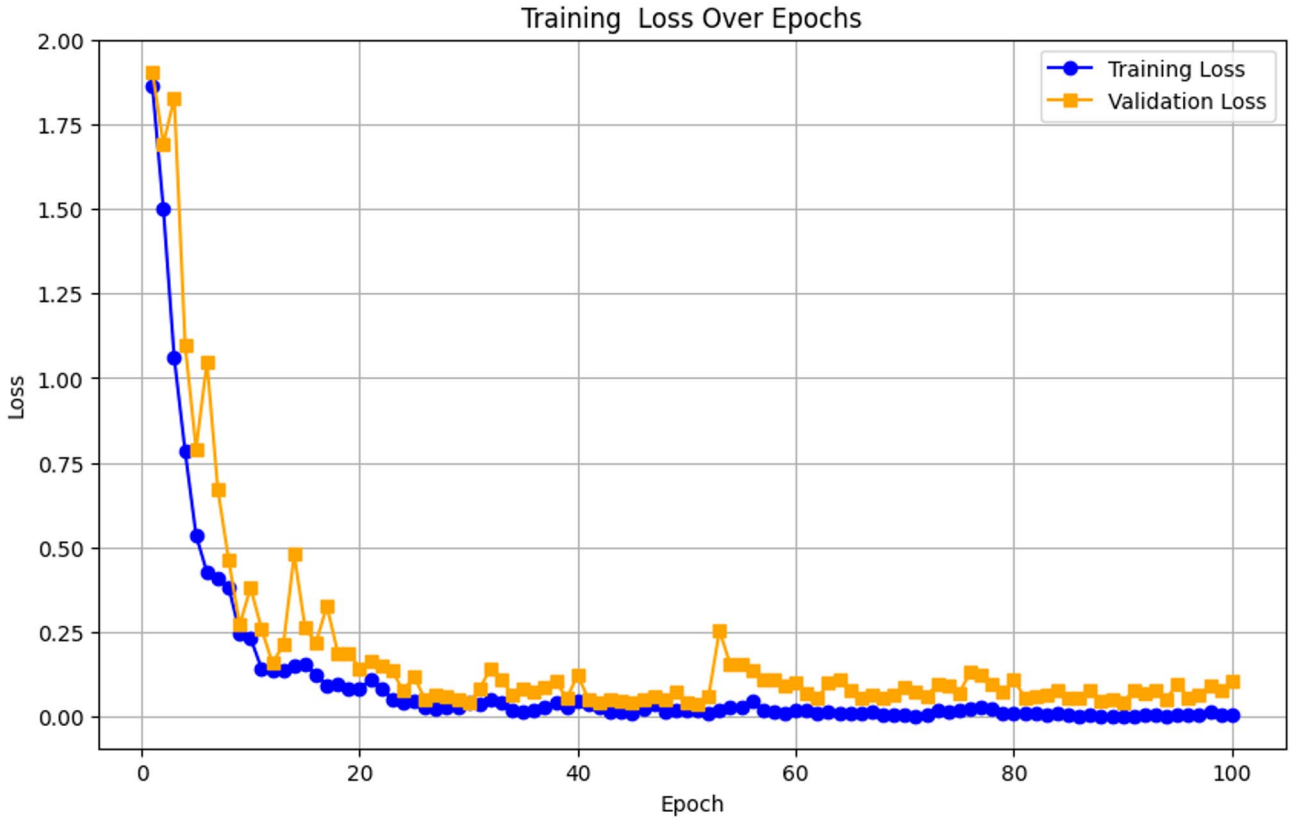


Fig. 11. Training and validation loss over epochs.

5% and 0.70 for 10%). The confusion matrix indicates that misclassifications predominantly occur between neighboring mid-range classes, such as 5% and 10% or 10% and 20%, implying inadequately differentiated thermal features at these concentrations.

A possible explanation for the model's deficiencies is the narrow range of adulteration levels, especially between 5% and 10%, and between 10% and 20%. The overlapping temperature characteristics of honey samples within these proximate adulteration ranges may impair the model's capacity to differentiate between them accurately. The percentages indicate intermediate adulteration levels, when variations in temperature distribution and texture patterns detected by thermal imaging are likely more modest, complicating classification.

The confusion matrix corroborates this problem, indicating that the majority of misclassifications transpire between these adjacent groups. For example, numerous 10% samples are inaccurately labeled as either 5% or 20%, and likewise, 5% samples are regularly misclassified as 3% or 10%. This overlap may also contribute to the noted declines in precision, recall, and F1 scores for these intermediate levels.

Another probable contributor to these constraints is the utilization of thermal pictures obtained from two devices with varying setups. Divergences in camera resolution, sensor sensitivity, and thermal calibration among the devices may result in data inconsistencies, complicating the model's ability to generalize and properly distinguish samples with minor thermal variations. The inconsistencies may have especially influenced the mid-range adulteration levels, where the thermal characteristics of honey are inherently less prominent and more vulnerable to noise from device variability.

Furthermore, the high AUC values (Fig. 14) reported above in Fig. 13 indicate the robust discriminatory ability of the model across all categories, with only slight deficiencies in the middle classes (5%, 10%, and 20%), where nuanced and overlapping thermal features complicate accurate classification. The extreme classes, including the pure (0%) and substantially adulterated (30%), as well as the extremely low adulteration levels (1% and 3%), exhibit unique thermal patterns that likely create unambiguous decision boundaries, allowing the algorithm to classify these instances with near-perfect accuracy. Conversely, the mid-range adulteration levels exhibit slightly inferior AUC scores, suggesting some overlap in thermal characteristics, a prevalent challenge in classification tasks involving incremental variations. At these intermediate concentrations, the thermal properties may not vary sufficiently, leading to occasional ambiguities in the model's classifications.

The PCA plot (Fig. 15 (a)) illustrates the separation of feature space using a linear transformation, hence capturing just linear correlations among the extracted features. Consequently, several classes, including Class 5 and Class 6, exhibit greater dispersion and overlap, indicating that their distinctions may be more intricate and nonlinear, which PCA fails to adequately represent. The hierarchical clustering dendrogram (Fig. 15 (b)) utilizes distance metrics and linkage criteria, accounting for both linear and nonlinear correlations, so offering a more adaptable method for grouping similar data. Clustering approaches depend solely on feature similarity in the

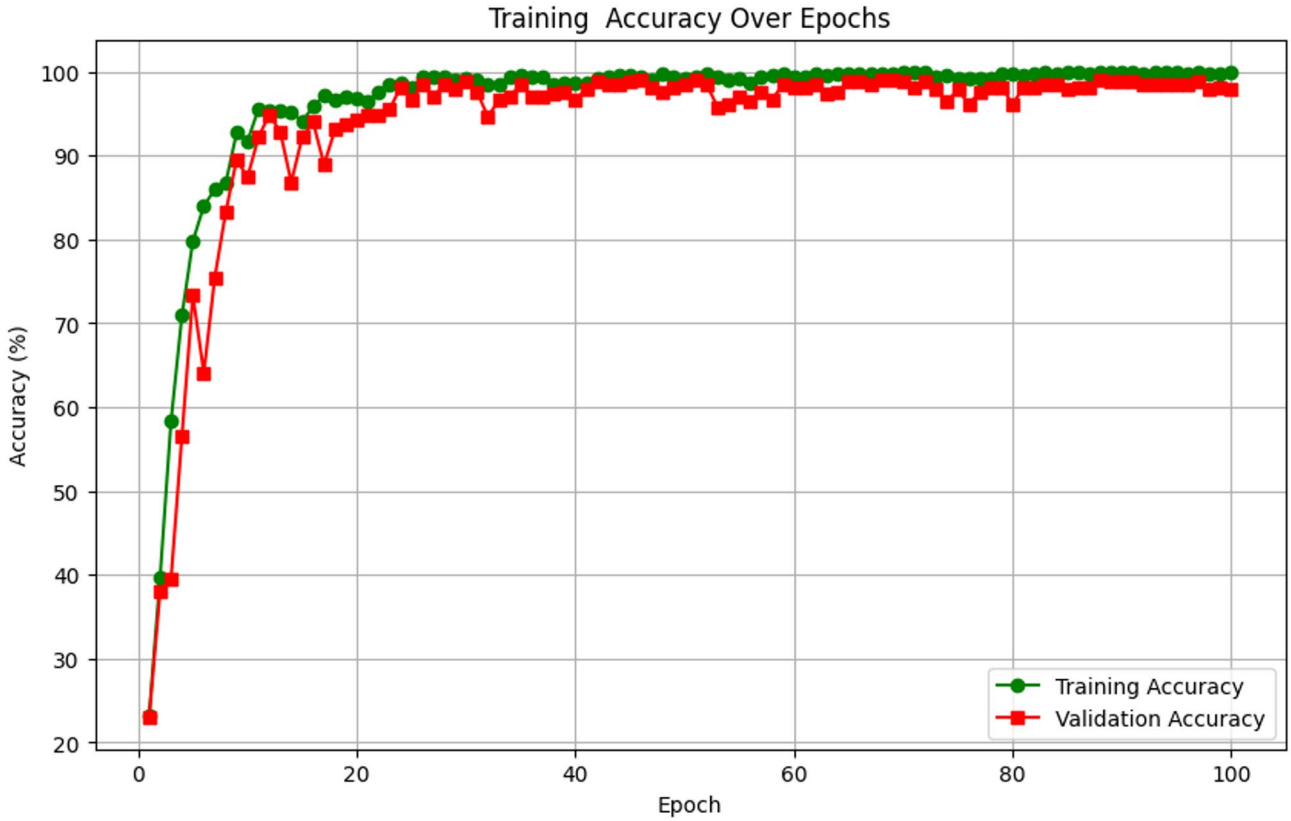


Fig. 12. Training and validation accuracy over epochs.

	0%	1%	3%	5%	10%	20%	30%
Precision	1.00	1.00	0.98	0.57	0.72	0.85	0.90
Specificity	1.00	1.00	1.00	0.97	0.96	0.92	0.99
Sensitivity (Recall)	0.94	0.82	0.90	0.63	0.68	0.91	0.93
F1 Score	0.97	0.90	0.93	0.60	0.70	0.88	0.92
Support	49	11	48	19	41	110	41

Table 7. Performance metrics of the model across various adulteration Levels.

absence of explicit class labels; thus, merging at greater distances suggests that certain classes possess shared feature properties. Nonetheless, the CBAM-enhanced RegNet, as a nonlinear deep learning model, catches intricate hierarchical patterns in the data, enabling it to acquire more discriminative representations than those revealed by PCA and hierarchical clustering.

Interpretability and regional attention in honey adulteration detection using Grad-CAM++Visualizations

Grad-CAM (Gradient-weighted Class Activation Mapping) is an effective method for visualizing and interpreting deep learning model predictions by emphasizing the most significant areas in an image that influence the model's classification[36,37]. The procedure commences by transmitting an input image over the network to acquire a preliminary class score. The gradients of this score are subsequently calculated in relation to the final convolutional feature maps, which include spatial information crucial for classification. The gradients are backpropagated, enabling the model to allocate importance values to various spatial locations according to their influence on the final choice. The resultant Grad-CAM heatmap is generated by integrating the activation maps with the corresponding significance values, thereby localizing the areas that significantly contributed to the classification.

The Grad-CAM heatmaps presented above (Fig. 16) elucidate the decision-making process of the CBAM-enhanced RegNet model in identifying honey samples with varying degrees of adulteration. The heatmaps delineate the areas in each honey image that significantly impacted class determination, enabling us to analyze how the program differentiates between pure and contaminated samples based on thermal patterns.

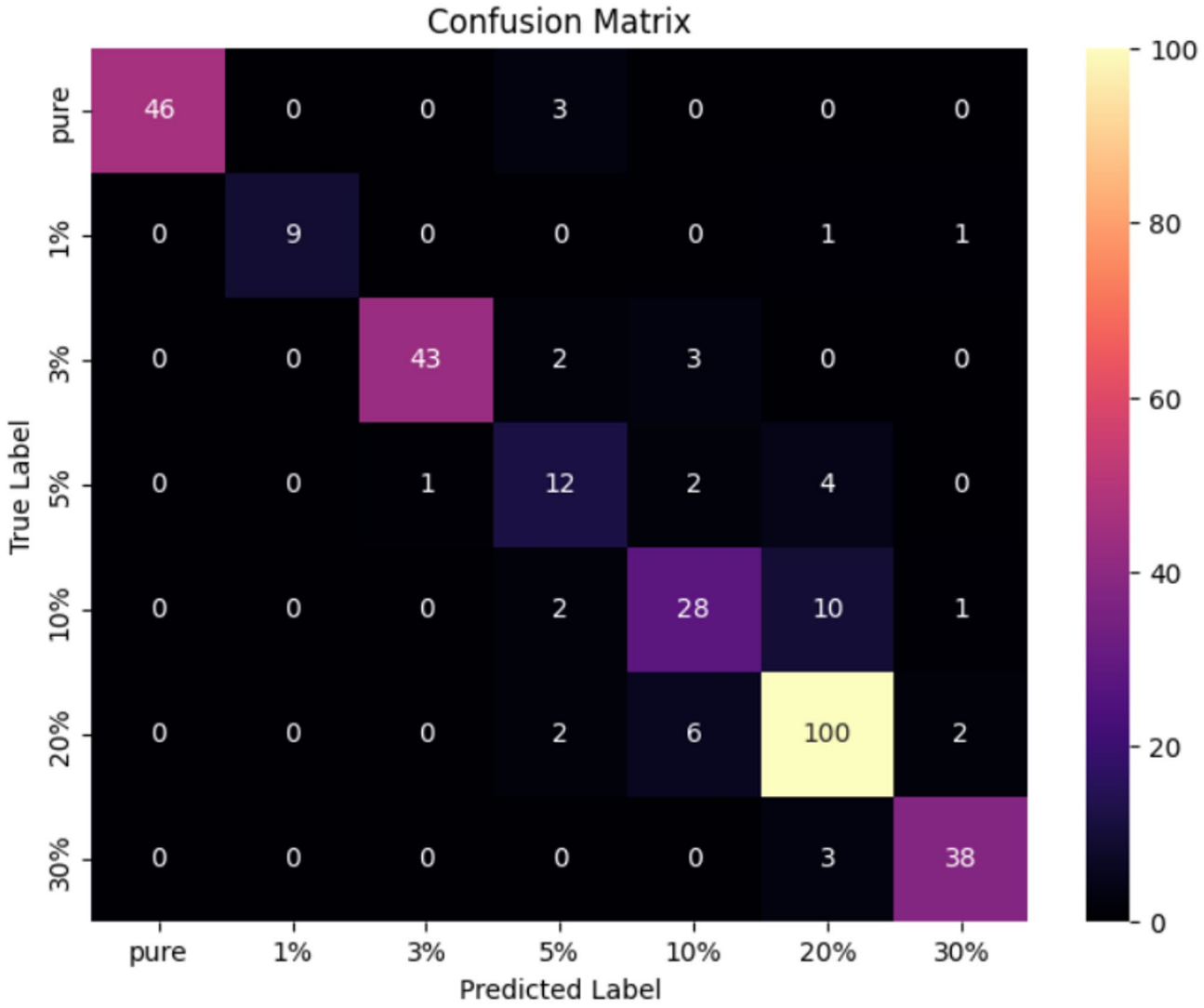


Fig. 13. Confusion matrix of the model.

For pure honey (Class 0), the heatmaps indicate equitable attention throughout core and peripheral regions, implying that the model discerns purity through uniform texture and thermal distribution. Conversely, at low adulteration levels (e.g., 1%), the model's attention becomes dispersed, signifying sensitivity to minor variations in thermal properties. As adulteration escalates (e.g., 5% - Class 3), the heatmaps reveal concentrated areas of interest, indicating that the model identifies preliminary indicators of contamination via distinct thermal fluctuations.

At elevated adulteration levels (e.g., 20% - Class 5), the heatmaps exhibit a marked concentration of focus in specific regions, corresponding to the significant temperature variations induced by the addition of glucose. This signifies that the model depends on these thermal markers as primary indicators of adulteration. Notably, even within identical classes, various honey samples display subtly distinct attention areas, illustrating the model's capacity to generalize across heterogeneous samples instead of only memorizing static patterns.

These findings validate that the CBAM-enhanced RegNet model, as a nonlinear deep learning architecture, proficiently captures hierarchical and discriminative features, rendering it superior to conventional feature extraction techniques such as PCA and hierarchical clustering. The Grad-CAM visualizations substantiate the model's proficiency in detecting thermal anomalies associated with honey adulteration, offering a clear and interpretable framework for quality control applications within the food industry.

Comparative analysis of our approach and previous studies

Our evaluation underscores significant distinctions between our approach and prior studies on honey adulteration identification (Table 8). Izquierdo et al.[13] employ thermal imaging with a CNN-based model; however, their dataset management raises issues regarding potential data leaking, as they fail to distinctly segregate training and testing sets. Their methodology entails utilizing a singular sample and subsequently partitioning the extracted images, which may lead to the presence of analogous images in both sets, possibly enhancing their reported

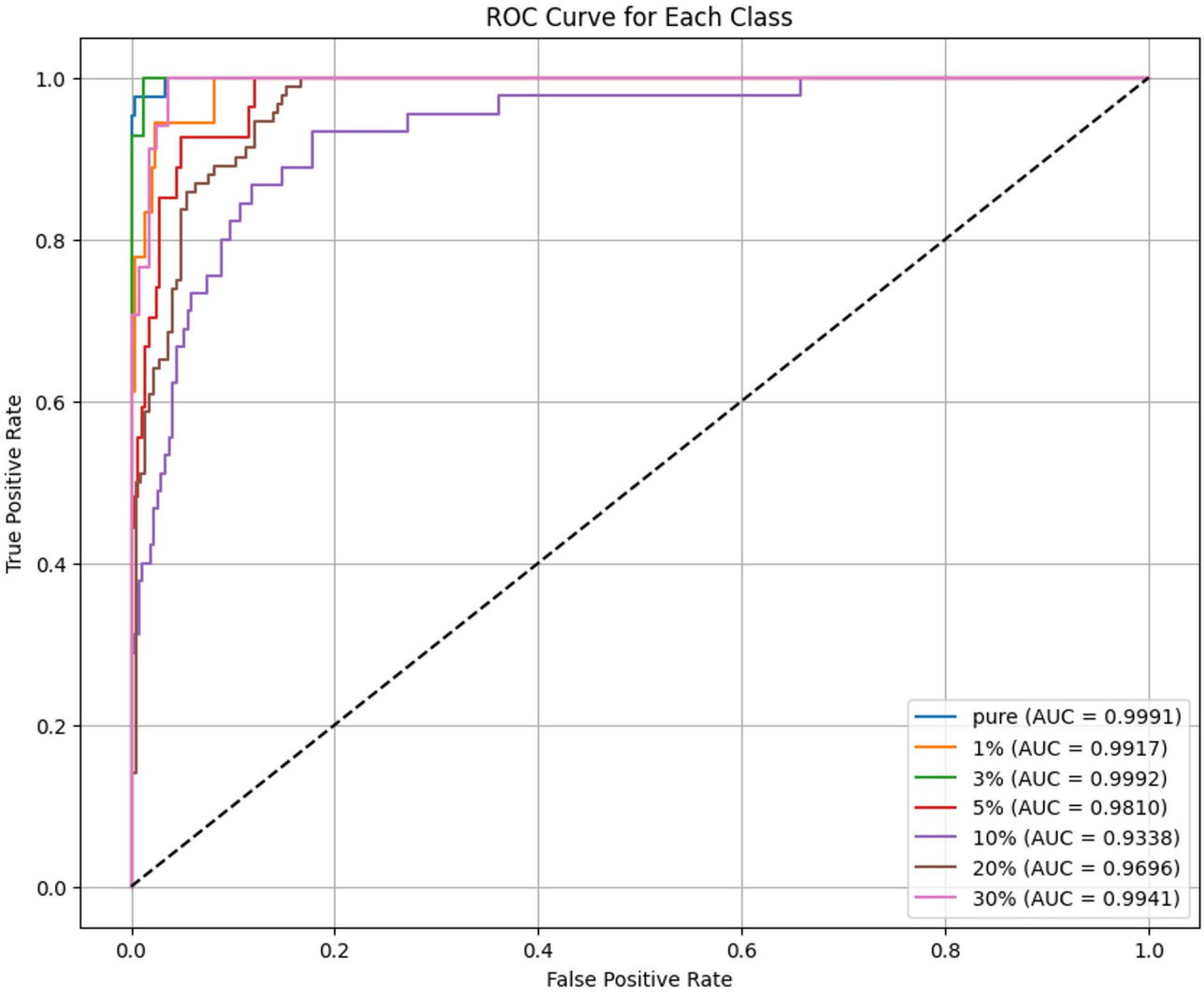


Fig. 14. ROC curves for multiclass classification.

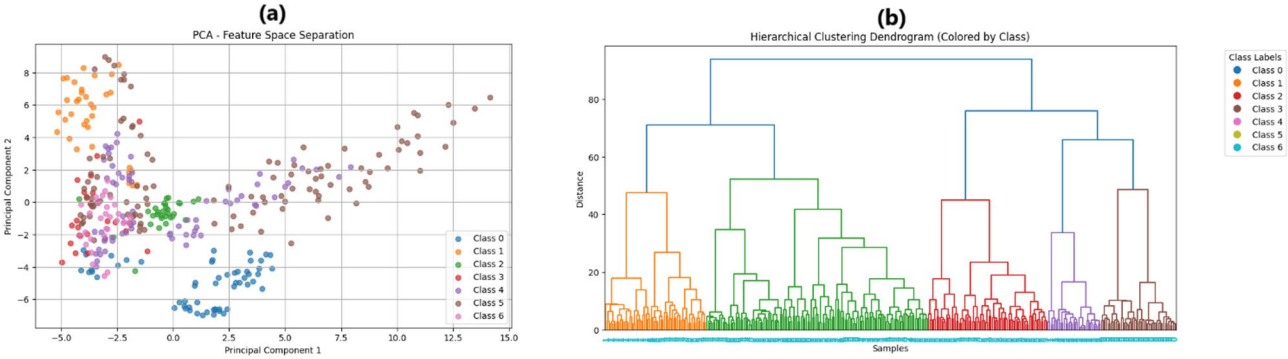


Fig. 15. PCA Projection and Hierarchical Clustering of Features Extracted by the CBAM-Enhanced RegNet Model.

accuracy of 0.93. Notwithstanding this, their model demonstrates a markedly inferior F1-score (0.63) and accuracy (0.48), indicating potential misclassification challenges.

Conversely, our methodology guarantees a stringent train-test division, preventing data leakage and attaining a balanced F1-score of 0.83 and enhanced precision of 0.85, so rendering it more dependable for practical applications. Ahmed et al.[38] and Geană et al.[39] utilize spectroscopy-based techniques, such as hyperspectral

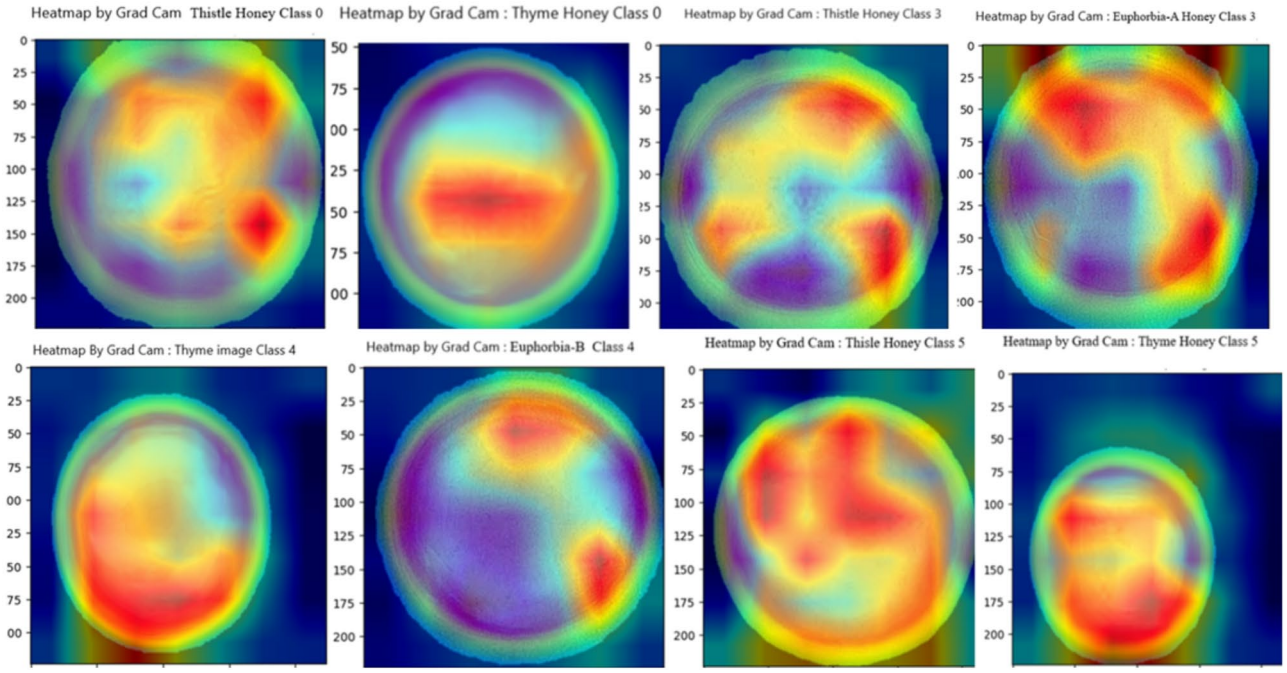


Fig. 16. Grad-CAM heatmaps of thermal images with varying adulteration levels.

References	Method	Accuracy	Recall	F1-Score	Precision	No. of Classes
Izquierdo, et al.[13]	Thermal Imaging + CNN	0.93	0.92	0.63	0.48	4
Ahmed, et al.[38]	Hyperspectral Imaging + ANN	0.98	0.99	0.98	0.98	5
Geană, et al.[39]	UV–Vis Spectroscopy + ANN	0.82	0.83	-	-	7
Our Approach	Thermal Imaging + CBAM_RegNet	0.85	0.83	0.83	0.85	7

Table 8. Comparison of our approach with previous studies on honey adulteration Detection.

and UV-Vis spectroscopy, which attain enhanced accuracy (up to 0.98). Nonetheless, these methodologies necessitate costly, specialist laboratory apparatus and proficient management, rendering them less attainable for extensive or field-based application.

Our approach, utilizing thermal imaging with CBAM_RegNet, offers a cost-effective and pragmatic option without compromising performance. While our accuracy (0.85) is marginally inferior to certain prior studies, our methodology guarantees superior dataset management and attains a commendable equilibrium between precision and recall, rendering it a resilient and economical option for honey adulteration detection.

Limitations

A significant limitation of this investigation is the absence of melissopalynological analysis to verify the botanical provenance of the unifloral honey samples. This method, regarded as the benchmark for floral source verification, proved impractical in our investigation due to insufficient specialist experience and equipment. To guarantee a satisfactory degree of authenticity, honey samples were obtained directly from cooperatives and beekeepers accredited by the National Office of Food Safety (ONSSA) in Morocco. These sources are recognized for generating unifloral honey varieties in areas with established floral predominance during harvest seasons. This method delivers a reliable indicator of botanical origin, but it does not substitute for the quantitative validation provided by melissopalynology. We recognize this as a limitation and suggest that future studies include melissopalynological testing to improve the validation of floral origin and for wider applicability of classification models.

Conclusion

Precisely distinguishing between genuine and contaminated honey is an essential component of apiculture. Current research introduces a novel model utilizing the attention mechanism to tackle this issue. This study analyzes five honey varieties: two thyme variants (Thyme-A and Thyme-B), two euphorbia variants (Euphorbia-A and Euphorbia-B), and one thistle variant, all of which are highly esteemed for their market demand. The program is engineered to identify adulteration levels between 0% and 30%, regardless of the honey's origin or variety. The results indicate substantial advancement in detecting adulteration levels, enhancing the quality control and verification of bee-derived goods.

One of the strengths of this research is the utilization of thermal images from different models of smartphones, which enhances the robustness of the model and its real-life usability. The splitting of samples into training and testing datasets prior to image capture also guaranteed that no images from one sample were included in both datasets. This enhances reliability by reducing variability in image capture. The results show that the combination of thermal imaging with a deep learning platform yields a robust, versatile, and multidimensional honey quality verification tool.

Potential opportunities

Expanding the model's applicability to encompass additional honey varieties and bee-derived goods, including royal jelly, is imperative. Subsequent investigations should evaluate the model's effectiveness in real-world processing contexts, which frequently exhibit greater complexity and unpredictability than controlled tests. Furthermore, modifying the model to identify supplementary types of adulteration would enhance its applicability. Iterative enhancements to improve accuracy and reduce errors will further increase its worth for the apiculture sector.

Data availability

The data that support the findings of this study are available from the corresponding author upon request.

Received: 16 July 2024; Accepted: 10 September 2025
Published online: 21 October 2025

References

1. Jamróz, M. K. et al. 1H and 13 C NMR-based sugar profiling with chemometric analysis and antioxidant activity of herbhoneys and honeys. *J. Sci. Food. Agric.* **94** (2), 246–255 (2014).
2. Leonard, R. J., Vergoz, V., Proschogo, N., McArthur, C. & Hochuli, D. F. Petrol exhaust pollution impairs honey bee learning and memory, Oikos, vol. 128, no. 2, pp. 264–273, Jan. (2019). https://doi.org/10.1111/oik.05405
3. Chen, Y. P. & Siede, R. Honey bee viruses. *Adv. Virus Res.* **70**, 33–80 (2007).
4. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., … Polosukhin,I. (2017). Attention is all you need. Advances in neural information processing systems, 30.
5. Devlin, J., Chang, M. W., Lee, K. & Toutanova, K. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*. (2018).
6. Zhu, Q. & Luo, J. Generative pre-trained transformer for design concept generation: an exploration. *Proc. Des. Soc.* **2**, 1825–1834 (2022).
7. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., … Amodei,D. (2020). Language models are few-shot learners. Advances in neural information processing systems, 33, 1877–1901.
8. Dosovitskiy, A. et al. Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*. (2020).
9. Touvron, H. et al. Training data-efficient image transformers & distillation through attention. In *International conference on machine learning* (pp. 10347–10357). PMLR. (2021), July.
10. Izquierdo, M. et al. Deep thermal imaging to compute the adulteration state of extra Virgin Olive oil. *Comput. Electron. Agric.* **171**, 105290 (2020).
11. Terrab, A., Recamales, A. F., Hernanz, D. & Heredia, F. J. Characterisation of Spanish thyme honeys by their physicochemical characteristics and mineral contents. *Food Chem.* **88** (4), 537–542 (2004).
12. https://www.flir.com/products/flir-one-pro/?model=435-0007-03.
13. Izquierdo, M. et al. Convolutional decoding of thermographic images to locate and quantify honey adulterations. *Talanta* **209**, 120500 (2020).
14. Rumelhart, D. E., Hinton, G. E. & Williams, R. J. Learning representations by back-propagating errors. *Nature* **323** (6088), 533–536 (1986).
15. Hochreiter, S. & Schmidhuber, J. Long short-term memory. *Neural Comput.* **9** (8), 1735–1780 (1997).
16. Samy Bengio Łukasz Kaiser. Can active memory replace attention? In Advances in Neural Information Processing Systems, (NIPS), (2016).
17. Uçar, A., Demir, Y. & Güzeliş, C. Object recognition and detection with deep learning for autonomous driving applications. *Simulation* **93** (9), 759–769 (2017).
18. Sharma, V., Gupta, M., Kumar, A. & Mishra, D. Video processing using deep learning techniques: A systematic literature review. *IEEE Access.* **9**, 139489–139507 (2021).
19. Sharma, S. & Guleria, K. Deep learning models for image classification: comparison and applications. In 2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE) (pp. 1733–1738). IEEE. (2022).
20. Vo, V. H., Sizikova, E., Schmid, C., Pérez, P. & Ponce, J. Large-scale unsupervised object discovery. *Adv. Neural. Inf. Process. Syst.* **34**, 16764–16778 (2021).
21. Desai, K. & Johnson, J. Virtex: Learning visual representations from textual annotations. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 11162–11173). (2021).
22. Minaee, S. et al. Image segmentation using deep learning: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **44** (7), 3523–3542 (2021).
23. Pang, G., Shen, C., Cao, L. & Hengel, A. V. D. Deep learning for anomaly detection: A review. *ACM Comput. Surv. (CSUR).* **54** (2), 1–38 (2021).
24. Benali Amjoud, A. & Amrouch, M. Convolutional neural networks backbones for object detection. In Image and Signal Processing: 9th International Conference, ICISP 2020, Marrakesh, Morocco, June 4–6, 2020, Proceedings 9 (pp. 282–289). Springer International Publishing. (2020).
25. Woo, S., Park, J., Lee, J. Y. & Kweon, I. S. Cbam: Convolutional block attention module. In Proceedings of the European conference on computer vision (ECCV) (pp. 3–19). (2018).
26. AOAC Official Method 980.23, Hydroxymethylfurfural in Honey: Spectrophotometric Method', in Dr. George W Latimer. Jr. (ed.), Official Methods of Analysis of AOAC INTERNATIONAL, 22nd Edition (New York, ; online edn, AOAC Publications, 4 Jan. 2023), (2023). https://doi.org/10.1093/9780197610145.003.3696
27. AOAC Official Method 958.09, Diastatic Activity of Honey', in Dr. George W Latimer. Jr. (ed.), Official Methods of Analysis of AOAC International, 22nd Edition (New York, ; online edn, AOAC Publications, 4 Jan. 2023), (2023). https://doi.org/10.1093/9780197610145.003.3701
28. https://www.gsmarena.com/samsung_galaxy_s21_fe_5g-10954.php
29. https://www.gsmarena.com/lg_velvet_5g-10209.php

30. https://www.ffmpeg.org/documentation.html
31. Tian, X., Wang, J. & Cui, S. Analysis of pork adulteration in minced mutton using electronic nose of metal oxide sensors. *J. Food Eng.* **119** (4), 744–749 (2013).
32. Ziou, D. & Tabbone, S. Edge detection techniques-an overview. Распознавание образов и анализ изображен/Pattern Recognition and Image Analysis: Advances in Mathematical Theory and Applications, 8(4), 537–559. (1998).
33. Xu, J. et al. RegNet: Self-regulated network for image classification. *IEEE Trans. Neural Networks Learn. Syst.* **34** (11), 9562–9567 (2022).
34. Bjorck, N., Gomes, C. P., Selman, B. & Weinberger, K. Q. Understanding batch normalization. Advances in neural information processing systems, 31. (2018).
35. Hsiao, T. Y., Chang, Y. C., Chou, H. H. & Chiu, C. T. Filter-based deep-compression with global average pooling for convolutional networks. *J. Syst. Architect.* **95**, 9–18 (2019).
36. Selvaraju, R. R. et al. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (pp. 618–626). (2017).
37. Chattopadhay, A., Sarkar, A., Howlader, P. & Balasubramanian, V. N. Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks. In 2018 IEEE winter conference on applications of computer vision (WACV) (pp. 839–847). IEEE. (2018), March.
38. Ahmed, E. Detection of honey adulteration using machine learning. *PLOS Digit. Health.* **3** (6), e0000536 (2024).
39. Geană, E. I., Isopescu, R., Ciucure, C. T., Gîjiu, C. L. & Joșceanu, A. M. Honey adulteration detection via Ultraviolet–Visible spectral investigation coupled with chemometric analysis. *Foods* **13** (22), 3630 (2024).

Author contributions
B.I: Conceptualization, Analyzed and interpreted the data, Wrote the paper.B.A: Conceptualization, supervised the findings of this work, reviewA.E : Conceptualization, Verified the analytical methodsD.S: Conceptualization, Verified the analytical methods, Analyzed and interpreted the data, supervised the findings of this work, Wrote the paper.D.H : supervised the findings of this work, review, and editing.

Funding
The authors are grateful for the funding of APRD Projects.

Declarations

Competing interests
The authors declare no competing interests.

Consent for publication
This article does not contain any studies with human participants performed by any of the authors.

Additional information
Correspondence and requests for materials should be addressed to S.D.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025